VICTORIA JUNIOR COLLEGE
BIOLOGY DEPARTMENT
JC2 PRELIMINARY EXAMINATIONS 2013
Higher 2

CANDIDATE NAME

CLASS

BIOLOGY 9648/01

Paper 1 Multiple Choice

27 September 2013

1 hour 15 minutes

Additional Materials: Multiple Choice Answer Booklet

READ THESE INSTRUCTIONS FIRST

Write in a soft pencil.
Do not use any staples, paper clips, highlighters, glue or correction fluid.
Write your name, Centre number and candidate number on the Answer Sheet in the spaces provided.

There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.
Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deduced for a wrong answer.
Any rough working should be done in this booklet

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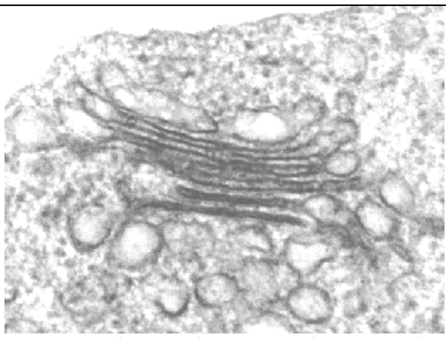
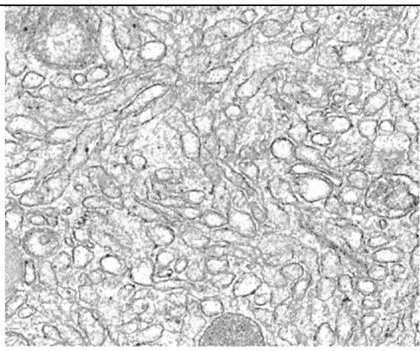
1 Below are electron micrographs of four different organelles that can be found in different eukaryotic cells. These micrographs are of *different* scale.



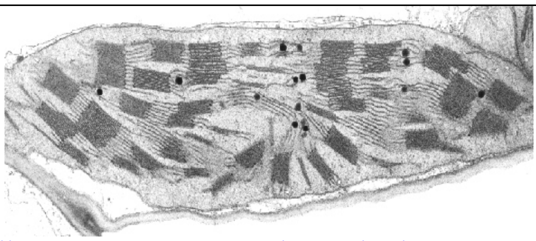
http://biology.unm.edu/ccouncil/Biology_124/Summaries/Cell.html

Organelle M



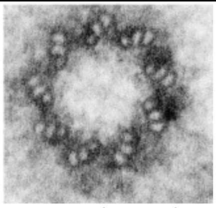
http://wiki.pingry.org/u/ap-biology/index.php/

Organelle N



http://www.biologie.uni-hamburg.de/b-online/e07/07a.htm

Organelle P



http://www.sciencedirect.com/science/article/pii/S0962892410001984

Organelle Q

Which of the following statements is true regarding the above organelles?

	Organelle M	Organelle N	Organelle P	Organelle Q
A	Is the sorting and packaging centre of the cell	Single membrane bound organelle responsible for lipid synthesis	Has a double membrane envelope, circular DNA and 70S ribosomes	Non-membrane bound organelle that is involved in the assembly of spindle fibres
B	Is connected to the rough endoplasmic reticulum so that proteins can be transported to it for packaging	Non-membrane bound organelle that is involved in protein synthesis	Is involved in the synthesis of ATP which is used for cellular activity	Membrane bound organelle involved in the organisation of spindle fibres
C	Double membrane bound organelle involved in modification of proteins	Is involved in the synthesis of carbohydrates	Extensive network of single membrane involved in the synthesis of proteins	Contains rRNA and proteins and are responsible for translation of proteins
D	Single membrane bound organelle responsible for lipid synthesis	Single membrane bound organelle involved in the digestion of food substances	Double membrane organelle containing euchromatin and heterochromatin	Is involved in the synthesis of kinetochore proteins

2 Lysosomal storage diseases are a group of approximately 50 rare inherited metabolic disorders that result from defects in lysosomal function.

Which of the following will be observed in the cells of someone suffering from a lysosomal storage disease?

A The accumulation of lipids due to the lack of the enzyme responsible for its hydrolysis.

B The accumulation of monosaccharides due to the lack of enzymes for carbohydrate biosynthesis.

C The accumulation of salts due to the inability to remove minerals.

D The overproduction of lysosomal membrane by the rough endoplasmic reticulum and Golgi apparatus.

3 A sample of yeast cells were grown in a culture. Radioactive amino acids were added to the solution in which they were being grown. At various times, samples of the cells were taken and the amount of radioactivity in different organelles was measured. The results are shown in the table.

Time after radioactive amino acids were added to the solution/ minutes	Amount of radioactivity present/arbitrary units		
	P	**Q**	**R**
1	21	120	6
20	42	68	6
40	86	39	8
60	76	28	15
90	50	27	28
120	38	26	56

Which best describes the identities of organelles **P**, **Q** and **R**?

	P	**Q**	**R**
A	Golgi apparatus	Rough endoplasmic reticulum	Secretory vesicles
B	Rough endoplasmic reticulum	Smooth endoplasmic reticulum	Golgi apparatus
C	Rough endoplasmic reticulum	nucleus	Golgi apparatus
D	Smooth endoplasmic reticulum	Golgi apparatus	Secretory vesicles

4 Which of the following statements is **false** of collagen and haemoglobin?

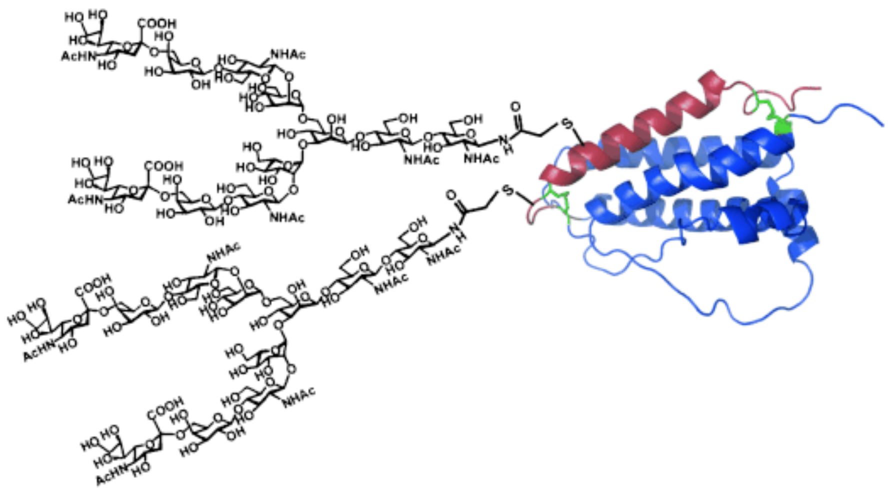
A Both are made up of amino acids.

B Both exhibit quaternary level of organisation.

C Collagen has a branched structure whereas haemoglobin has a globular structure.

D There is less variability in the monomers present in collagen compared to haemoglobin.

5 The figure below shows the structure of a biomolecule extracted from a cell.



Below are some statements regarding the structure, property and function of biomolecules with structures similar to that shown above. Which of the following statements are true?

(i) The biomolecule has both hydrophilic and hydrophobic properties.

(ii) It is important for cell to cell adhesion.

(iii) The synthesis of this molecule requires only the rough endoplasmic reticulum.

(iv) When completely hydrolysed, all the monomers of this biomolecule are soluble in water.

A **(i)** and **(iii)** only

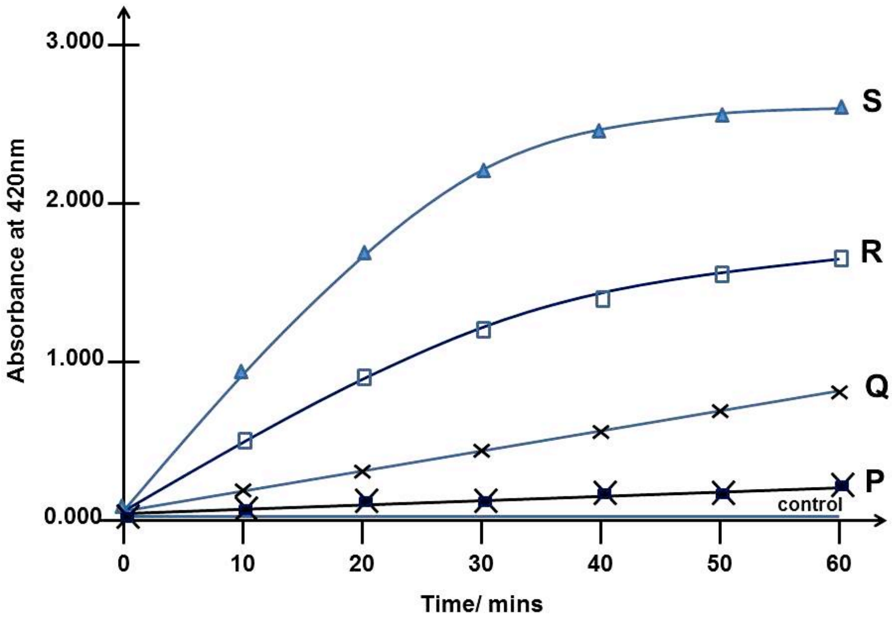
B **(ii)** and **(iv)** only

C **(i)**, **(ii)** and **(iv)** only

D **(ii)**, **(iii)** and **(iv)** only

6 The enzyme β-galactosidase plays an important role in cellular metabolism by breaking down lactose into glucose and galactose. The enzyme kinetics of β-galactosidase can be determined by using a lactose analog, o-nitrophenyl-β-galactopyranoside (ONPG) which is cleaved by β-galactosidase to o-nitrophenol (ONP).

ONPG is colourless while the reaction product ONP is a yellow compound. The rate of ONP formation is proportional to the absorbance value. The absorbance values of samples containing the ONPG and ß-galactosidase enzyme were determined at 10 minute intervals for a total of 60 minutes. All the results shown below are obtained by varying only one factor.



Based on the figure above, identify the correct conclusion(s).

(i) The graphs **P** and **Q** show that an increase in concentration of ONPG leads to an increase in the rate of product formation.

(ii) The graph **S** confirms that ß-galactosidase is denatured by high temperature.

(iii) The graph **S** shows that beyond 50 minutes, all the substrate have been used up.

(iv) The graphs **P**, **Q** and **R** show the effect of competitive inhibition when ONPG concentration is in excess.

(v) The control is a flat line because boiled and cooled ß-galactosidase is unable to catalyse the breakdown of ONPG to ONP.

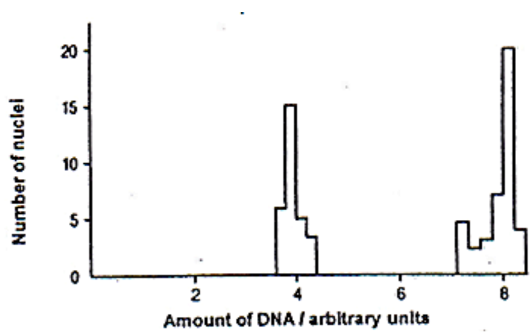
A **(ii)** only

B **(iii)** only

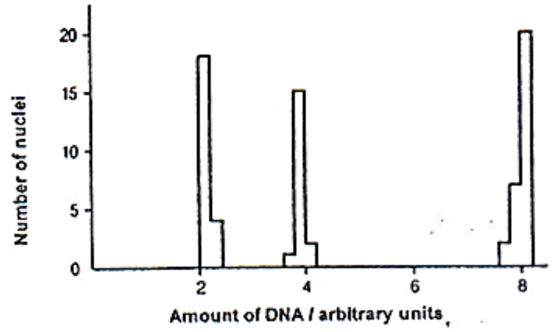
C **(i)** and **(iv)** only

D **(i)**, **(iii)** and **(v)** only

7 The graphs below show the amount of DNA in the nuclei of cells taken from different parts of a mammalian testis.



Graph 1



Graph 2

Which correctly describes one **unique** event taking place in the cells from graph **1** and graph **2** respectively?

	Graph 1	**Graph 2**
A	Duplication of DNA	Separation of homologous chromosomes
B	Separation of identical sister chromatids	Separation of non-identical sister chromatids
C	Separation of sister chromatids	Formation of gametes
D	Breaking and rejoining of homologous regions of chromosomes	Separation of homologous chromosomes

8 *Staphylococcus aureus*, a bacterium, has a genome size of 2.8 X 10^6 base pairs. During DNA replication, 400 base pairs are formed per second at one replication fork.

How long will it take to replicate the entire *Staphylococcus aureus* genome?

A 29 minutes

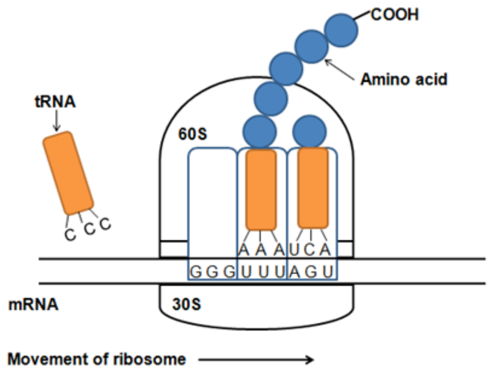
B 58 minutes

C 116 minutes

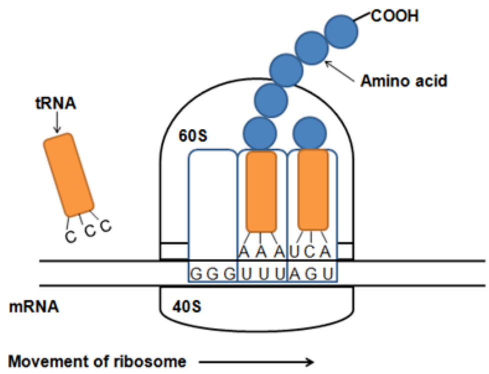
D 233 minutes

9 Which of the following diagrams show the correct process of eukaryotic translation?

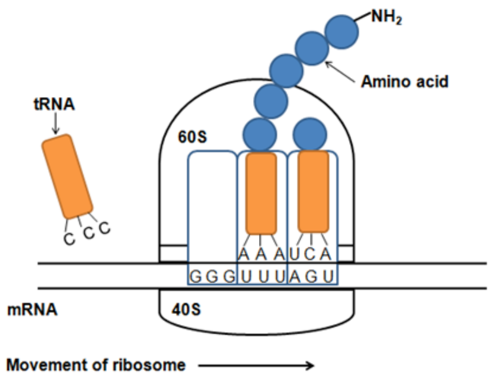
A



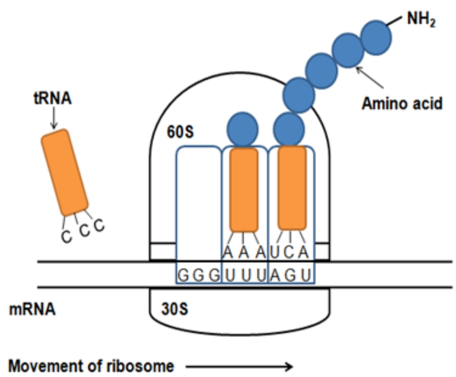
B



C



D



10 Fabry's Disease is a disease that results from a mutation that occurs in the *α-galactosidase A* gene.

The sequence of part of the normal and mutated alleles for *α-galactosidase A* gene is shown below.

Normal allele

Codon	37	38	39	40	41	42	43	44	45
mRNA	CCU	UGG	ACC	CAG	AGG	UUC	UAA	GGC	GGA

Mutated allele

Codon	37	38	39	40	41	42	43	44	45
mRNA	CCU	UGG	ACC	CCG	CAG	AGG	UUC	UAA	GGA

Using the information of the normal and mutated alleles above, it is reasonable to conclude that

A a frame shift mutation has occurred.

B a duplication has occurred.

C an insertion of an amino acid has occurred in the mRNA.

D the polypeptide that is translated from the mutated allele will be longer.

11 Alternative splicing is a regulated process in gene expression that results in a single gene coding for multiple proteins.

Which of the following statement(s) is/ are true?

(i) Alternative splicing produces different mRNAs in different tissues as a result of specific transcription factors binding to enhancers and silencers.

(ii) Proteins translated from alternatively spliced mRNAs will have different amino acid sequence and perform different biological functions

(iii) Alternative splicing is regulated by a system of specialized proteins that bind to splice sites flanking the exons on the DNA sequence

(iv) Alternative splicing is regulated by a system of specialized proteins that bind to splice sites on the pre-mRNA

A **(i)** only

B **(iii)** only

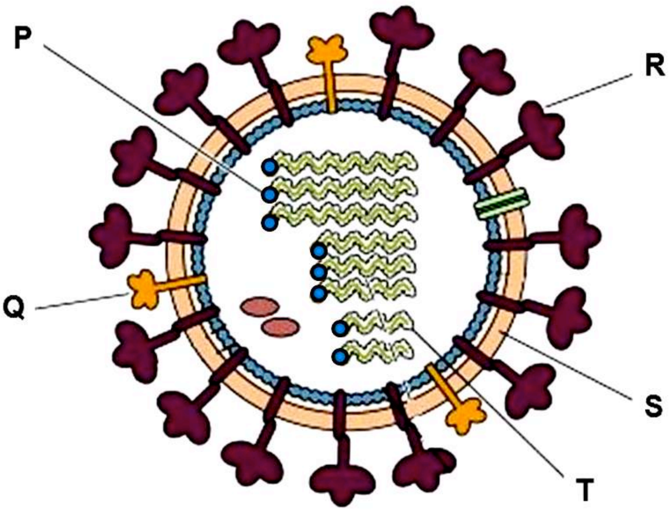
C **(ii)** and **(iii)** only

D **(ii)** and **(iv)** only

12 Which option below describes correctly a similarity and a difference between transduction and conjugation?

	Similarity between Transduction and Conjugation	**Difference between**	
		Transduction	**Conjugation**
A	They involve vertical transmission.	Transduction involves the lysis of the donor bacterial cell.	Conjugation does not result in the lysis of cells.
B	They combine DNA from two bacterial cells into genome of one.	Genes from the bacterial genome is transferred.	Genes from F plasmid is transferred.
C	Bacteria acquire new genetic material.	Bacterial cell takes up foreign DNA from culture medium.	Double strand of F plasmid transferred from one bacterium to another.
D	They involve horizontal gene transfer.	DNA is transferred by bacteriophage from one bacterium to another.	Single strand of F plasmid is transferred from one bacterium to another.

13 The diagram is shows the structure of an influenza virus.



Which of the following statements concerning the lettered components are correct?

(i) Mutations that disrupt the function of **P** will result in the inability of the virus to initiate infection in the host cell.

(ii) Influenza viruses are able to cause pandemics in humans due to changes in **Q** and **R** by a slight mutation.

(iii) **Q** and **R** are unlikely targets for vaccination because they undergo mutation constantly.

(iv) The symptoms associated with influenza infection are due directly to loss of epithelial cells as the new viruses obtain **S** during budding.

(v) The host cell enzymes are not required to form the complementary RNA from **T**.

A **(i)** and **(v)**

B **(i)**, **(ii)** and **(v)**

C **(ii)**, **(iii)** and **(iv)**

D **(iii)**, **(iv)** and **(v)**

14 Regulatory proteins can control the expression of genes in prokaryotes by

A binding to the DNA in an active state to inhibit transcription

B binding to the DNA in the presence of an inducer molecule to allow transcription

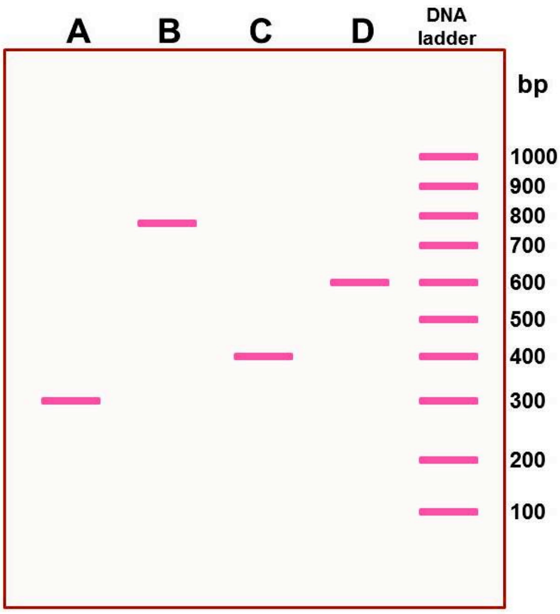
C binding to the DNA to upregulate transcription unless an active repressor is present

D All of the above

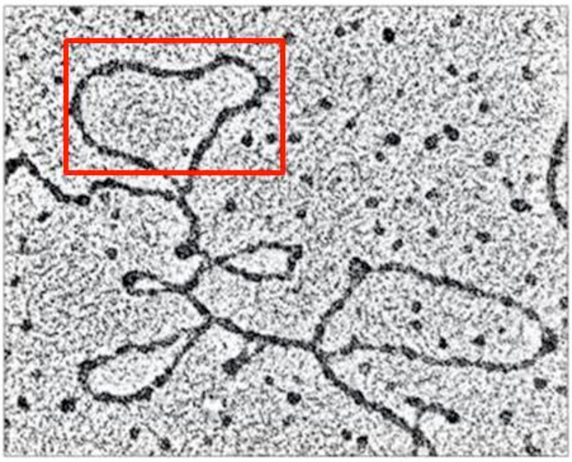
15 Gene **X** contains three exons and two introns and codes for a polypeptide containing 99 amino acids. Intron 1 is 250 base pairs long while Intron 2 is 150 base pairs long. The mRNA of Gene **X** has a 5'UTR of 100 ribonucleotides, a 3' UTR of 50 ribonucleotides as well as a string of 150 adenine ribonucleotides.

After reverse transcription, the double-stranded cDNA was subjected to gel electrophoresis.

Identify the lane with the correct band fragment in the figure below.



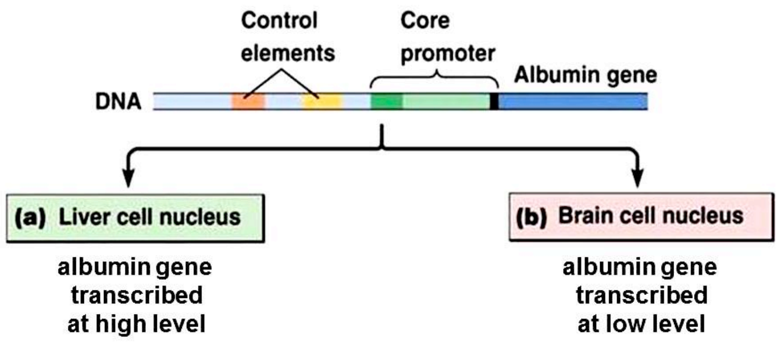
16 The figure below is a photograph showing structures that are formed when mature mRNA is added to a pool of single-stranded DNA. Some DNA remains single stranded while others form hybrid structures.



Identify the single-stranded structure boxed up above.

A Exons **B** Introns **C** Repetitive DNA **D** Control elements

17 The gene for the protein albumin is associated with two control elements, as well as the promoter. These control elements are recognised by regulatory transcription factors which bind and facilitate transcription of the albumin gene at a high level.



Differential albumin gene expression in liver cells and brain cells result from

A Liver and brain cells contain RNA polymerase and the general transcription factors but differ in the regulatory elements in the albumin gene.

B Brain cells lack RNA polymerase and the general transcription factors resulting in low transcription of the albumin gene.

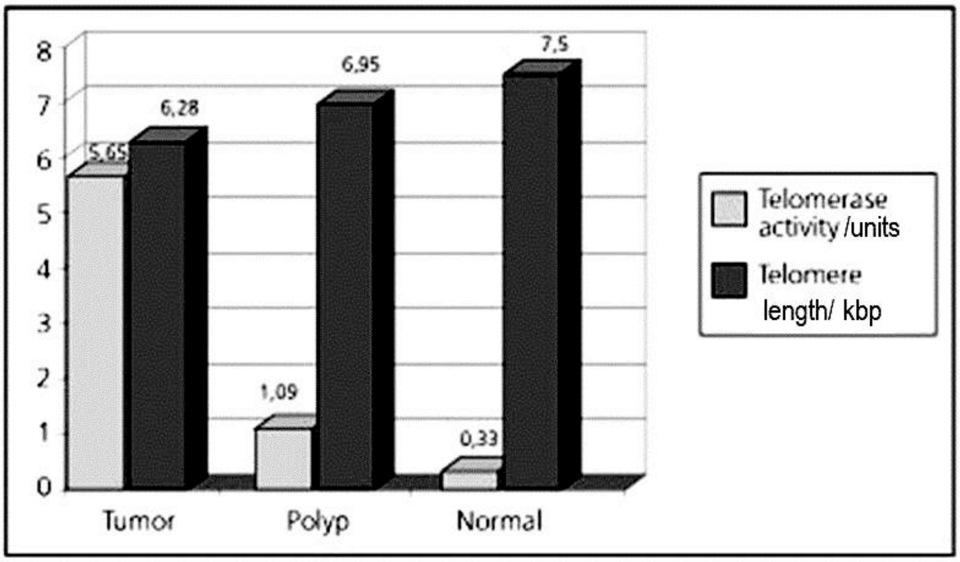
C Brain cells do not contain the regulatory transcription factors that are required to bind the control elements that promote the assembly of transcription complex at the promoter.

D Albumin gene that is not required for expression in the brain cell acquires somatic mutations in the core promoter so that transcription initiation complex can no longer assemble efficiently.

18 Which of the following shows correctly the types of genetic changes that are **least** likely to be found in an oncogene and a tumor suppressor gene of tumor cells?

	Oncogene	**Tumor suppressor gene**
A	Gene amplification	Chromosomal deletion
B	Chromosomal deletion	Substitution mutation
C	Substitution mutation	Chromosomal translocation
D	Mutation resulting in a stop codon	Gene amplification

19 In familial adenomatous polyposis (FAP), numerous polyps form in the epithelium of the large intestine and if left untreated, these benign polyps develop into colon cancer. To investigate the role of telomerase in the cancer process, telomerase activity was measured in different tissue samples. At the same time, telomere length determination in these samples was performed by Southern blot. The results are shown below.



Using your knowledge of the function of telomerase, what conclusion(s) can be reasonably inferred from the figure above?

(i) Increase in telomerase activity is likely to be responsible for the maintenance of telomere length in the development of cancer.

(ii) Telomerase activity is needed to maintain telomere length in normal epithelial cells.

(iii) Since normal epithelial cells have long telomeres, they will continue to divide.

(iv) Drugs that inhibit telomerase activity can still be effective against polyp cells although they already possess long telomeres.

A **(i)** and **(iv)** only

B **(ii)** and **(iii)** only

C **(i)**, **(ii)** and **(iii)** only

D **(i)**, (**iii)** and **(iv)** only

20 In guinea pigs, the allele for rough coat is dominant over the allele for smooth coat and the allele for black fur is dominant over the allele for white fur. The genes for fur colour and texture are not linked.

Two guinea pigs which are heterozygous for both fur colour and texture were mated together. What is the probability that the offspring would be homozygous for one of the dominant alleles?

A 1 in 2

B 1 in 3

C 3 in 8

D 2 in 9

21 Coat colour in rabbits is controlled by a gene with four alleles. The order of dominance for these alleles is as follows:

agouti (**C**) > chinchilla ($\mathbf{C^c}$) > himalayan ($\mathbf{C^h}$) > albino (**c**)

What is the maximum number of classes of coat colour obtained from a cross between an agouti rabbit and a Himalayan rabbit?

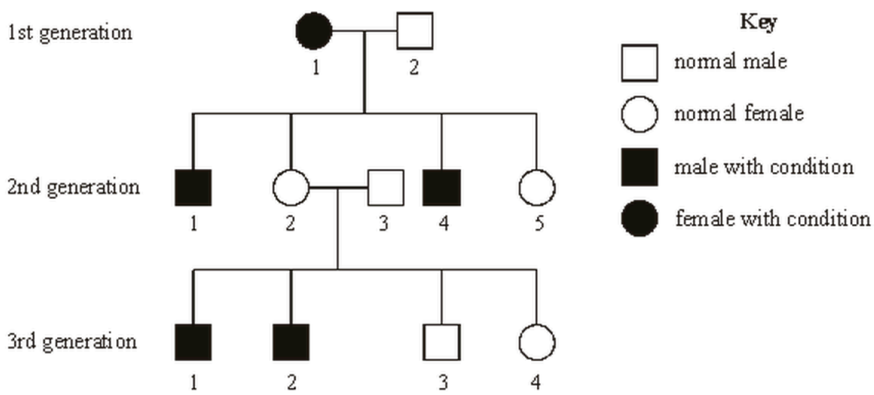
A 1

B 2

C 3

D 4

22 The diagram below is the pedigree of a family with a disease condition.



If individual 4 from the 3rd generation marries a normal male, what is the probability that the first offspring is a normal male?

A 1/4 **B** 1/3 **C** 1/2 **D** 3/8

23 Familial hypercholesterolaemia (FH) increases the risk of premature coronary artery disease.

Fig. 23a shows the pedigree of a family with members A to H on whom DNA analysis was performed (**Fig. 23b**). Individual **H** has high cholesterol levels due to the inheritance of a mutant allele of the LDL receptor gene.

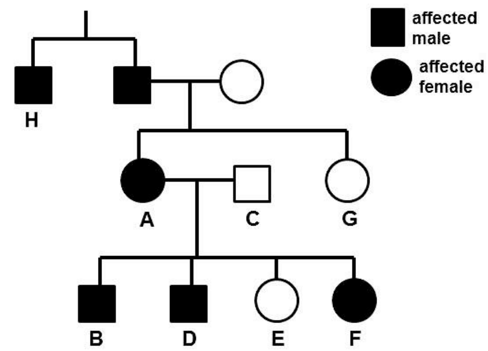


Fig. 23a

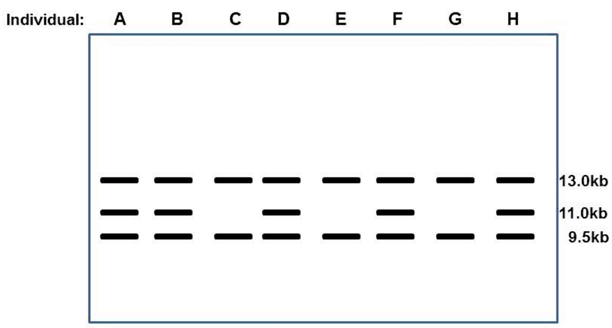


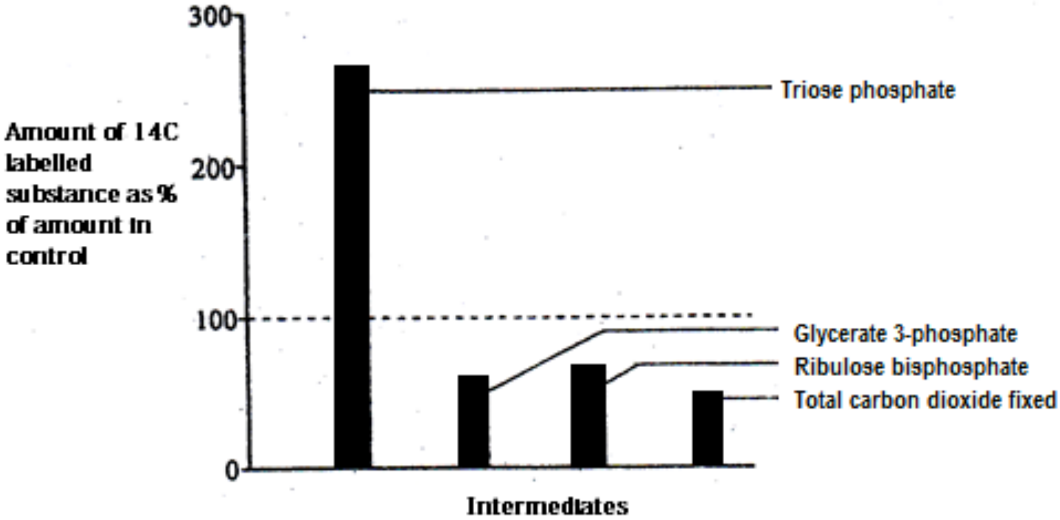
Fig. 23b

Based on **Fig. 23a** and **Fig. 23b**, what conclusions can be correctly drawn?

A	FH is inherited in an autosomal dominant manner.	The 11.0kb band is the result of the mutant allele.
B	FH is inherited in an autosomal dominant manner.	The 13.0kb band is the result of the mutant allele.
C	FH is inherited in a sex-linked recessive manner.	The 13.0kb and 9.5kb bands are the result of the mutant allele.
D	FH is inherited in a sex-linked recessive manner.	The 11.0kb band is the result of the mutant allele.

24 An experiment was conducted to test the properties of a chemical **G** on the photosynthetic capabilities of a unicellular agar, Chlorella. An illuminated suspension of the alga was treated with carbon dioxide labeled with ^{14}C in the presence of an unknown chemical **G**. The light was switched off and the amount of radioactivity present in some intermediates was determined after 10 minutes in the dark.

A control suspension of alga without chemical **G** being added was treated in exactly the same manner. The bar chart below shows the amount of radioactivity in these intermediates in the alga with chemical **G** added as a percentage of the intermediates in the control alga.



Which option correctly describes the action of chemical **G**?

A **G** binds to NADPH produced in light reactions and prevent its oxidation process.

B **G** competes with triose phosphate for the active site of the enzyme that converts triose phosphate into hexose phosphate.

C **G** prevents the regeneration of ribulose bisphosphate at the stage after triose phosphate was formed.

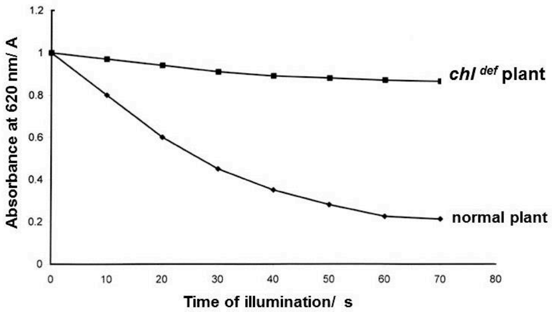
D **G** inhibits the carboxylase enzyme, preventing carbon fixation from taking place efficiently.

25 Dichlorophenol-indophenol (DCPIP) is used to investigate the light-dependent stage of photosynthesis. DCPIP is a blue dye which acts as an electron acceptor. As DCPIP is reduced, its blue colour disappears. The depletion of DCPIP is indicated by a drop in absorbance.

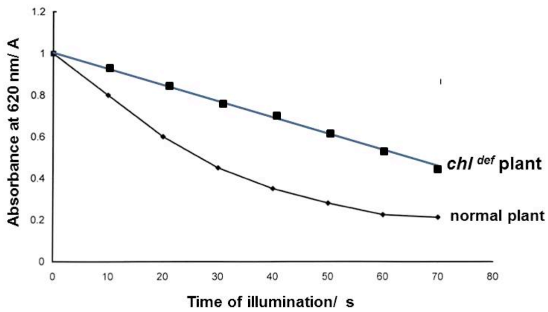
Leaf extracts were prepared from plants with normal *chlorophyll a* gene and plant that contain a deletion mutation in *chlorophyll b* gene (*chl*^{def}), thus reducing the amount of chlorophyll b present. These extracts were mixed with DCPIP solution and exposed to white light for different lengths of time and absorbance were determined.

Identify the correct set of results.

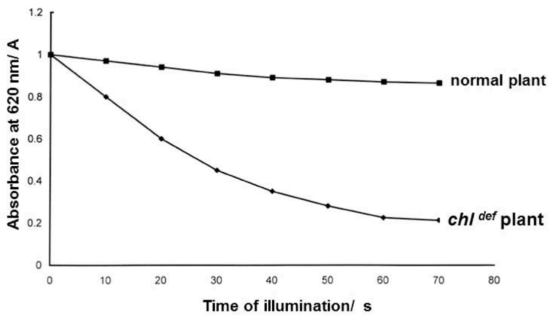
A



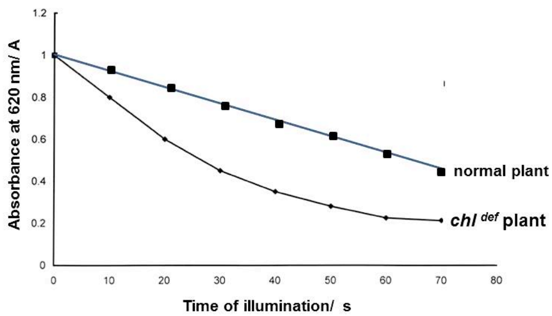
B



C



D



26 In an experiment, four tubes were set up as shown in the table below.

tube	contents
1	Glucose + homogenized animal cells
2	Glucose + mitochondria
3	Glucose + cytoplasm lacking organelles
4	Pyruvate + homogenized animal cells

If all other conditions are kept constant, which of the following shows the amount of ATP produced in each tube in **increasing** order?

A **1 – 3 – 4 – 2**

B **2 – 3 – 4 – 1**

C **4 – 2 – 3 – 1**

D **3 – 2 – 1 – 4**

27 From yeast cells, a mixture of enzymes can be obtained that catalyzes the conversion of glucose to carbon dioxide and ethanol.

Which combination of substances would increase the rate of ethanol production by these enzymes?

A ATP and lactate

B Reduced NAD and coenzyme A

C ATP and reduced NAD

D NAD and carbon dioxide

28 A person suffers from a rare genetic disease which causes the beta cells in the Islets of Langerhans to degenerate prematurely. This person ate a carbohydrate meal and four hours later, went to exercise vigorously for an hour. What will be the concentration of glucose, insulin and glucagon in his blood after the exercise?

	glucose	insulin	glucagon
A	high	low	low
B	high	high	low
C	low	low	high
D	low	low	low

29 Which of the following is a similarity between G-protein-linked receptors and tyrosine-kinase receptors?

A Both have hydrophobic stretches of amino acids spanning across the plasma membrane several times that serve to anchor the receptors.

B Both are directly involved in carrying out enzymatic reactions to initiate signal transduction processes.

C Activation of both receptors eventually lead to an amplification of the signal via their individual signal transduction pathway.

D Both receptors undergo conformational change through dimerization when bound by extracellular ligands.

30 Which of the following accounts for the presence of a refractory period following an action potential?

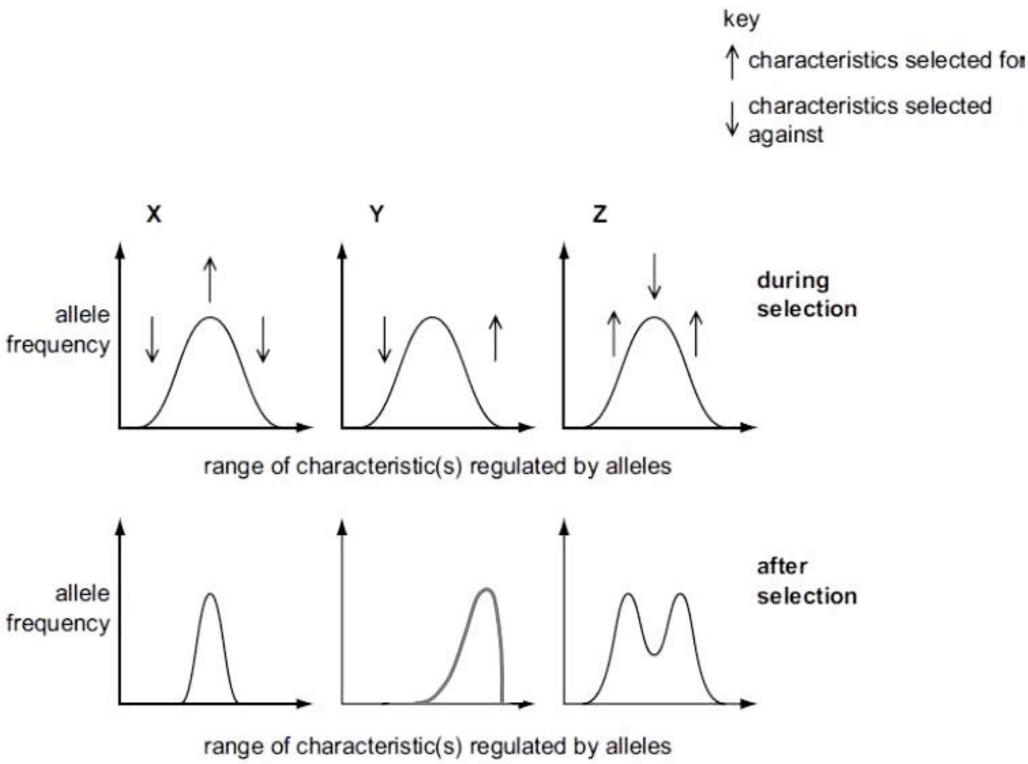
A The leak channels for the sodium ion are closed and thus sodium ions cannot diffuse into the axon to cause another depolarisation.

B Both the leak channels for sodium and potassium ions are closed and hence the resting potential cannot be re-established for the next action potential to be generated.

C Voltage-gated potassium ion channels open slowly during repolarisation and remained open for a period of time after the action potential.

D Voltage-gated sodium ion channels are inactivated and these are unable to reopen again immediately upon depolarisation.

31 The graphs represent the frequency of alleles in species **X**, **Y** and **Z** during and after selection.



In which species has evolution taken place?

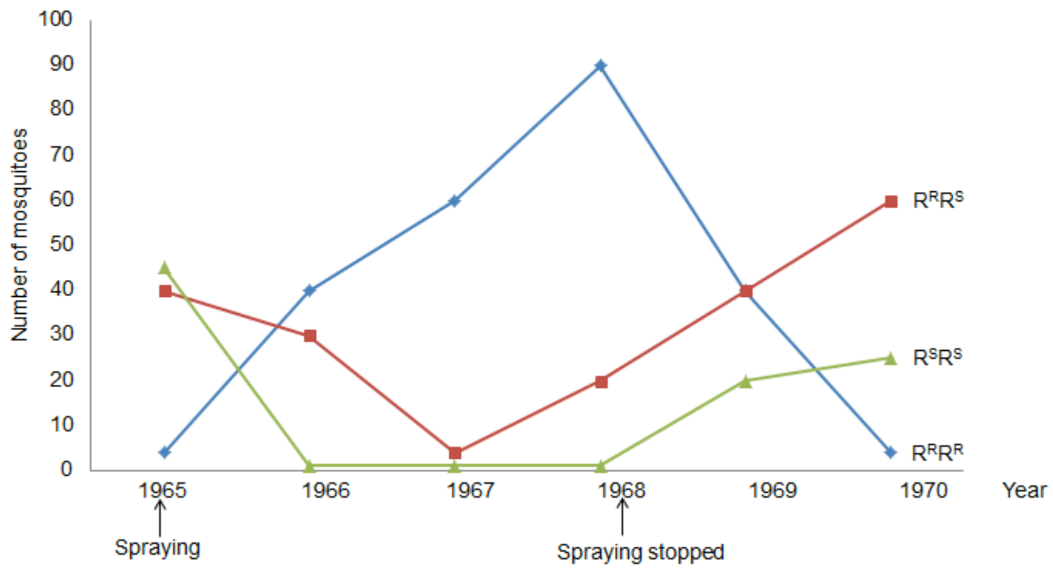
A **X** only

B **Y** only

C **Y** and **Z** only

D None of the above

32 In the mosquito, there is a gene locus which has two alleles, R^R = Resistance and R^S= Sensitive, which are involved in resistance, in particular to DDT, the insecticide. The graph below shows the number of mosquitoes of the three phenotypes (and genotypes), collected in 1965, when DDT was first used, through to 1970, two years after the spraying of DDT stopped.



From the data it is possible to conclude that

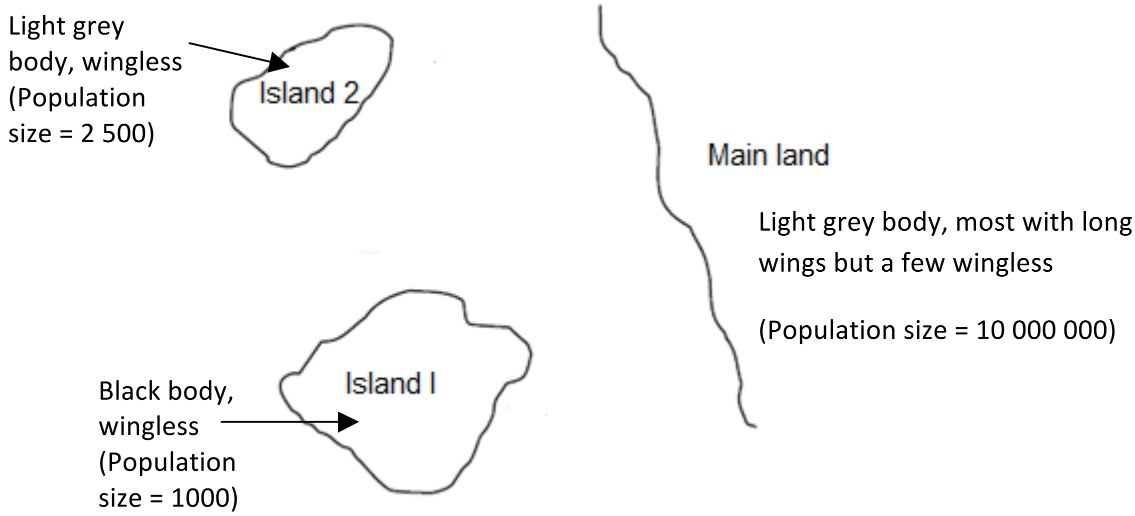
A the frequency of the R^S allele is greater than the frequency of the R^R allele in 1968.

B many generations after the removal of DDT the R^R allele would disappear from the population.

C after the removal of DDT from the environment in 1968, having the R^RR^R genotype reduces the chance of survival.

D between 1967 and 1968 in the presence of DDT in the environment, the mosquitoes with the R^RR^S genotype are the most likely to survive.

33 The diagram below shows the distribution of an insect species in the main land and two nearby islands.



Which of the following **cannot** be inferred from the information above?

A The soil in Island 1 may be darker in colour.

B The population in Island 2 may be founded by a few light grey bodied insects without wings that were blown there.

C Convergent evolution in the two islands has resulted in the presence of the wingless phenotype.

D The insect population on the mainland has more genetic variability than those in the two islands.

34 Several closely related frog species of the genus *Rana* are found in the forests of the South-eastern United States. The species boundaries are maintained by reproductive barriers. The statements below describe the different type of reproductive barriers that exist between the different species.

I Males of one species sing only when its predators are absent; males of another species sing only when its predators are present.

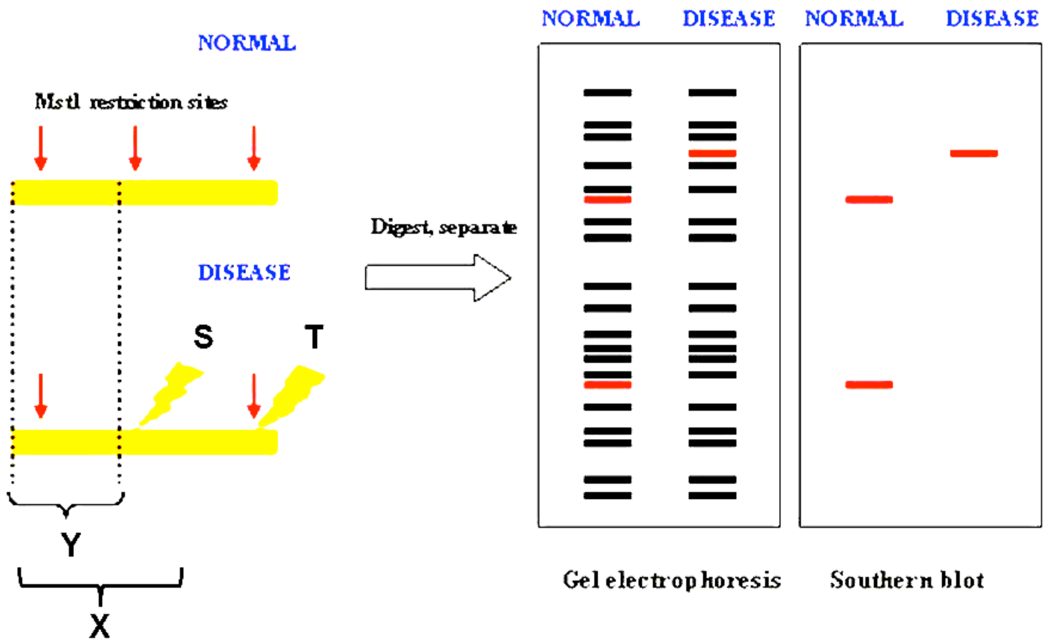
II One species mates at the season when daylight is increasing from 13 hours to 13 hours, 15 minutes; another species mates at the season when daylight is increasing from 14 hours to 14 hours, 15 minutes.

III Two species of frogs belonging to the same genus occasionally mate, but the offspring fail to develop and hatch.

Which of the following matches correctly the statement to the type of reproductive barrier that is described?

	Statement I	**Statement II**	**Statement III**
A	temporal	behavioural	hybrid sterility
B	seasonal	geographical	hybrid sterility
C	sympatric	temporal	gametic incompatibility
D	behavioral	temporal	hybrid inviability

35 A difference in homologous DNA sequences can be detected by the presence of fragments of different lengths after digestion of DNA samples with specific restriction endonucleases and using a labelled DNA sequence that hybridizes with one or more fragments of the digested DNA sample after they were separated by gel electrophoresis, thus revealing a unique blotting pattern, as shown in the figure below.



Which combination of probe and mutation correctly explains the band patterns in the Southern blot?

	Probe used for hybridization to digested fragment	Location of point mutation in DNA sequence in the disease
A	X	T
B	Y	T
C	X	S
D	Y	S

36 Which of the following DNA sequences would a restriction enzyme usually recognize and cut?

A 5' – ATGCAC – 3'
3' – TACGTG – 5'

B 5' - GATATC – 3'
3' – CTATAG – 5'

C 5' - TAGATA – 3'
3' – ATCTAT – 5'

D 5' - AATATA – 3'
3' – TTATAT – 5'

37 Plant physiologists attempted to produce papaya plants using tissue culture. They investigated the effects of different concentrations of plant growth factors on small pieces of the stem tip from a papaya plant. Their results are shown in the table below.

Concentration of auxin / $\mu mol\ dm^{-3}$	Concentration of cytokinin / $\mu mol\ dm^{-3}$		
	5	25	50
0	No effect	No effect	Leaves produced
1	No effect	Leaves produced	Leaves produced
5	No effect	Leaves produced	Leaves and some plantlets produced
10	Callus produced	Leaves and some plantlets produced	Plantlets produced
15	Callus produced	Callus and some leaves produced	Callus and some leaves produced

What evidence from the table supports that cells from the stem tip are totipotent?

A Under certain concentrations of cytokinin to auxin, there is production of both leaves and calluses or leaves and plantlets showing that more than one type of cells can be produced at any one time.

B Stem tip cells are able to produce calluses when the concentration of cytokinin to auxin is in the ratio of 1:2 or 1:3.

C Stem tip cells are able to produce leaves although they are supposed to differentiate into stems.

D Stem tip cells are able to give rise to plantlets.

38 Marker genes are often inserted into genetically engineered crop plant cells, along with desired genes. Bacterial antibiotic resistance genes are sometimes used as marker genes. These may include short DNA repeats to make them unstable so that they are quite quickly eliminated by the genetically engineered crop plant cells.

Which is **not** a reason why elimination of such marker genes is favoured?

A It is theoretically possible for the antibiotic resistance marker gene in human food to pass to bacteria in the human gut.

B It is difficult to carry out repeated transformations using the same antibiotic.

C The antibiotics may affect the growth and differentiation of the fields of crop plants.

D There are only a few such antibiotic resistance marker genes available.

39 Which of the following is a reason why gene therapy involving the delivery of a normal allele for proto-oncogene is not likely to be successful in the treatment of cancer?

A The normal allele will code for protein products that will allow the formation of a tumor.

B For a particular proto-oncogene, there are many alleles controlling the production of the normal protein and hence one allele is insufficient.

C The delivery of the normal allele is likely to cause the cell cycle to stop, hence the stem cells will not be able to divide and differentiate to give more cells.

D The normal allele for proto-oncogene is recessive and hence will not be expressed in the presence of the dominant mutated allele.

40 A proposed treatment for a rare hereditary recessive disease involved the use of stem cells that were modified via gene therapy. After stem cells modified with a copy of the functional dominant allele were introduced into the body of the patients, symptoms continued to surface.

Which of the following is/are **not** a reason/reasons for the failure of the treatment?

I The disease can be attributed to the effects of a number of genes located on different chromosomes.
II Lymphocytes of the patients recognize stem cells as foreign and target them for destruction.
III Plasticity of the stem cells were limited and unable to differentiate into the specific cell types required to carry out the function.
IV Inefficient system of delivery prevented the stem cells from reaching the site of disease.

A **I** only

B **I** and **II** only

C **II** and **III** only

D **I**, **III** and **IV** only